Contents

pmsimstats2025: Comprehensive Status and Path Forward

Replicating Hendrickson et al. (2020) with modern practices and addressing the carryover-power problem

2026-03-18

1. Project Goal

pmsimstats2025 aims to:

1. **Faithfully replicate** the Hendrickson et al. (2020) N-of-1 trial simulation using modern R practices (tidyverse, parallel processing, Docker/renv, pre-validated sigma matrices)
2. **Diagnose** why power falls as carryover half-life increases in the original simulation
3. **Demonstrate** that explicitly modeling carryover in the analysis recovers lost power (the two-scenario comparison)

The project maintains three sibling repositories: the original published codebase (pmsimstats), the modernized rewrite (pmsimstats2025), and a pedagogical simplification (pmsimstats-simple).

2. What Works

pmsimstats2025 has made substantial architectural improvements over the original:

- **Reproducible environment**: Docker + renv + Makefile (the original had none)
- **Parallel processing**: `furrr::future_map_dfr()` across iterations (the original was sequential)
- **Sigma caching**: Pre-built and validated sigma matrices avoid redundant computation
- **Two-stage MVN generation**: Conditional draw using pre-computed `Sigma_22_inv` and `cond_mean_transform`
- **Pre-simulation validation**: Eigenvalue checking and condition number reporting for all sigma structures
- **Two-scenario comparison framework**: Runs each condition with and without carryover in the analysis model, directly quantifying the cost of ignoring carryover
- **Extensive documentation**: 40+ documents covering mathematical foundations, Hendrickson alignment, carryover theory, and implementation decisions

The binary treatment assignment vectors (the `ondrug` paths) match exactly between the original and pmsimstats2025 for all four designs (OL, OL+BDC, Crossover, Hybrid). The expectancy parameters also match.

3. Critical Divergences Identified

The code review identified five critical differences between pmsimstats2025 and the original that produce non-comparable power estimates. These are listed in order of impact.

3.1 Biomarker-Treatment Interaction Mechanism

The single most consequential difference.

The original encodes the biomarker-treatment interaction entirely within the MVN covariance matrix. There is no biomarker term in the mean structure. The mechanism:

- On-drug timepoints: `corr(bm, BR) = c.bm`
- Off-drug timepoints: `corr(bm, BR)` scaled by `mm1/mm0` (Ron Thomas logic) or set to 0

- Participants with higher biomarker values tend to have higher BR draws purely through multivariate correlation
- The interaction emerges stochastically from `mvrnorm()`

pmsimstats2025 encodes the interaction in the mean structure:

```
effective_BR_rate = BR_rate * (1 + bm_mod * bm_centered)
```

This is a deterministic modulation: high-biomarker participants get a larger BR mean regardless of the covariance structure. The sigma matrix uses a constant `c.bm` across all timepoints, with no treatment-dependent adjustment.

Prior documentation (`why-post-mvn-adjustment-analysis.md`, `hendrickson-vs-our-detailed-comparison.md`) identified that an earlier version of pmsimstats2025 applied the biomarker effect three times (correlation + population mean shift + post-MVN adjustment), double-counting the interaction. The recommendation to use "Option 1: pure correlation" was made but not yet fully implemented in `simulation_clustered.R`.

Impact: The mean-based mechanism produces stronger, more detectable interactions per participant than the correlation-based mechanism. Power estimates are systematically higher in pmsimstats2025 than the original for the same nominal parameter values.

3.2 Per-Path vs. Shared Sigma Generation

The original generates data **separately for each trial path** using a **path-specific sigma matrix**. Each path's covariance structure depends on its treatment sequence because the Ron Thomas correlation scaling adjusts biomarker-BR correlations based on whether each timepoint is on-drug or off-drug for that specific path.

pmsimstats2025 generates all participants from a **single cached sigma matrix per design**. Path-specific treatment effects are applied after the MVN draw, in the mean structure. The covariance structure is identical for all paths.

Impact: In the original, Path A (drug-first) and Path B (placebo-first) in a crossover have different covariance matrices. The biomarker correlates with BR at different timepoints depending on path. In pmsimstats2025, the biomarker correlates identically with BR regardless of path.

3.3 Time Since Discontinuation (`tsd`)

The original computes `tsd` as cumulative off-drug interval widths:

```
tsd[i] = t_wk[i] * (od[i] != 1)
# then accumulate: tsd[i] += tsd[i-1] when off-drug
```

At the first off-drug timepoint, `tsd` equals the measurement interval duration (1-4 weeks).

pmsimstats2025 computes `tsd` as calendar time since transition:

```
tsd = week - discontinuation_week
```

At the first off-drug timepoint, `tsd = 0` (same week as the transition event). This produces a systematic offset of one measurement interval at every off-drug timepoint.

Detailed path-by-path comparison (`tsd_comparison.md`):

```
Hybrid Path A:
  Week:    4   8   9  10  11  12  16  20
  orig:    0   0   0   0   1   2   0   4
  main:    0   0   0   0   0   1   0   0
```

```
CO Path A:
  Week:  2.5   5  7.5  10  12.5  15  17.5  20
  orig:    0   0    0   0   2.5  5.0  7.5  10.0
  main:    0   0    0   0   0    2.5  5.0   7.5
```

Impact: At the first off-drug measurement, pmsimstats2025 computes `carryover_effect = (1/2)^0 = 1` (full retention), while the original computes `(1/2)^(2*1/0.2) = 0.001` (0.1% retention). This is a 1000x difference in carryover at the most informative contrast point. The `predictor_comparison.tex` and `tsd_comparison.md` documents thoroughly characterize this as a step-function power cliff.

3.4 Carryover Scale Factor

The original uses `scalefactor = 2` in data generation:

```
brmeans[p] += brmeans[p-1] * (1/2)^(2 * tsd / t1half)
```

pmsimstats2025's `simulation_clustered.R` omits the scale factor:

```
carryover_effect = (1/2)^(tsd / carryover_t_half)
```

(Note: `pm_functions.R` retains `scale_factor = 2` as default, but the primary simulation script does not use it.)

Impact: The effective half-life is doubled in pmsimstats2025 relative to the original for the same nominal parameter. Combined with the tsd offset, this compounds to produce dramatically different carryover behavior.

Additionally, `hendrickson-carryover-analysis.md` discovered that the original uses `scalefactor = 2` in data generation but `carryover_scalefactor = 1` in the analysis model. This DGP-analysis mismatch is a feature of Hendrickson's design (carryover is intentionally unmodeled in the analysis), but the different scale factors add an additional layer of misspecification.

3.5 Cumulative Drug Exposure (`tod` / `weeks_on_drug`)

The original computes `tod` as cumulative time-weighted exposure in weeks, with gap reset:

```
Hybrid Path A:
  Week:  4   8   9  10  11  12  16  20
  tod:   4   8   9  10   0   0   4   0
```

At week 16 (drug re-initiated after off-drug gap), `tod = 4` (just the new 4-week interval), not 14 (total lifetime exposure). The Gompertz restarts from its steep early phase.

pmsimstats2025 uses `cumsum(treatment)` (a count of on-drug measurement occasions) that plateaus off-drug and continues from the accumulated count when drug resumes:

```
Hybrid Path A:
  Week:  4   8   9  10  11  12  16  20
  wod:   1   2   3   4   4   4   5   5
```

Impact: Different units (weeks vs. count), different off-drug behavior (0 vs. plateau), and different re-initiation behavior (reset vs. continue). These feed into different growth models (Gompertz vs. linear), further amplifying the divergence.

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4. Additional Divergences

These are less impactful but should be addressed for full fidelity.

4.1 Response Growth Model

The original uses a modified Gompertz function (sigmoidal, saturating) with four parameters per factor (max, displacement, rate, sd). pmsimstats2025 uses a linear rate model (unbounded, one parameter per factor). For short trials with small rate parameters, these may be approximately equivalent, but they produce different dose-response shapes.

4.2 Autocorrelation Structure

The original uses compound symmetry within each factor (constant correlation regardless of time lag). pmsimstats2025 uses time-based AR(1) where correlation decays with elapsed time: `rho^|t_i - t_j|`. This is a deliberate improvement in realism.

4.3 ER SD Scaling

The original scales expectancy response standard deviations by the expectancy parameter (blinded timepoints have halved ER variance). pmsimstats2025 does not scale ER SDs; expectancy enters only through the mean.

4.4 BM-BR Correlation at Timepoint 1

The original's Ron Thomas loop starts at `p > 1`, so the first timepoint's BR has zero correlation with the biomarker even when on-drug. pmsimstats2025 applies `c.bm` to all timepoints including the first.

4.5 BM-ER and BM-TR Correlations

The original correlates the biomarker only with BR. pmsimstats2025 also correlates the biomarker with ER and TR at half-strength (`c.bm * 0.5`). The original has no such cross-component biomarker correlations.

4.6 Correlation Parameter Values

The original uses `c.br = c.pb = c.tv = 0.8`, `c.cf1t = 0.2`, `c.cfct = 0.1`. pmsimstats2025 reduced these to `0.75`, `0.12`, and `0.05` for improved positive definiteness conditioning.

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5. The Power Dropoff Problem

5.1 The observation

Hendrickson's published results show power declining as carryover half-life increases from 0 to 0.1 to 0.2 weeks. The two-scenario comparison in pmsimstats2025 is designed to demonstrate that this decline can be mitigated by explicitly modeling carryover in the analysis.

5.2 Why does power drop?

Hendrickson's analysis model does not include carryover. The `lme_analysis.R` defaults to `op$carryover_t1half = 0`, which makes `Dbc` a binary on/off indicator. When carryover exists in the data but not in the model:

1. Off-drug observations carry residual drug effect
2. The binary `Dbc` codes them as 0 (no drug)
3. The mismatch between actual and modeled drug exposure inflates residual variance
4. Standard errors on the interaction coefficient increase
5. Power declines

5.3 Scale factor compounds the problem

The data generation uses `scalefactor = 2` but the analysis uses `scalefactor = 1`. Even if the analysis included carryover, the decay rates would not match. At `tsd = 1, t1half = 0.2`:

- DGP retention: `(1/2)^(2*1/0.2) = (1/2)^10 = 0.001`
- Analysis retention: `(1/2)^(1*1/0.2) = (1/2)^5 = 0.031`
- 30x mismatch

This is a separate, unmodeled confound on top of the binary vs. continuous `Dbc` issue.

5.4 Candidate mechanisms (post-tsd-fix)

After fixing the tsd computation to match the original, if excessive power dropoff persists, three mechanisms should be investigated:

1. **Scale factor mismatch**: Testable by running with matched scale factors (both 1 or both 2)
2. **PD enforcement distortion**: Small nonzero correlations at off-drug timepoints may cause `make.positive.definite()` to perturb the on-drug correlations differently than exact-zero correlations
3. **Cascading carryover in BR means**: The recursive formula `brmeans[p] += brmeans[p-1] * decay` means each off-drug timepoint's carryover depends on the previous, creating nonlinear contamination patterns

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6. The Two-Scenario Comparison

This is pmsimstats2025's primary methodological contribution.

6.1 Design

For each parameter combination:

- **Scenario 1** (WITH carryover modeling): Analysis uses `effective_drug_weeks` which accounts for carryover decay. The analysis model matches the DGP.
- **Scenario 2** (WITHOUT carryover modeling): Analysis uses `weeks_on_drug` (binary, no decay). This replicates Hendrickson's approach and should reproduce the published power decline.

6.2 Expected outcome

```
carryover = 0: Scenario 1 = Scenario 2 (no confound)
carryover > 0: Scenario 1 > Scenario 2 (modeling helps)
```

The gap between the two scenarios quantifies the recoverable power – the cost of ignoring carryover that Hendrickson's approach pays.

6.3 Current status

The framework is implemented in `simulation_clustered.R` but cannot produce valid results until the five critical divergences (Section 3) are resolved. The comparison is meaningful only when Scenario 2 reproduces the original's power curves.

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7. Changes Required

Phase 1: Faithful Replication (Critical)

These changes bring pmsimstats2025's DGP into alignment with the original. Until these are complete, power estimates are not comparable.

Change 1: Correlation-based biomarker interaction. Remove the mean-structure biomarker modulation (`BR_rate * (1 + bm_mod * bm_centered)`) and implement the Ron Thomas differential correlation logic in the sigma matrix. The biomarker-BR correlation must be `c.bm` at on-drug timepoints and scaled/zero at off-drug timepoints. This requires:

- Building path-specific sigma matrices (Change 2)
- Computing BR means via Gompertz of `tod` (Change 5)
- Applying the Ron Thomas `mm1/mm0` scaling

Change 2: Per-path sigma generation. Replace the single cached sigma per design with path-specific sigma matrices. Each path's sigma must reflect its treatment sequence in the biomarker-BR correlation block. This may reduce the effectiveness of sigma caching (more structures to pre-build) but is essential for the correlation-based interaction to work.

Change 3: Fix tsd computation. Replace the calendar-subtraction approach with interval-based accumulation matching the original:

```
last_on_drug_week = if_else(
  treatment == 1, week, NA_real_)
last_on_drug_week = zoo::na.locf(
  last_on_drug_week, na.rm = FALSE)
tsd = if_else(
  treatment == 0 & !is.na(last_on_drug_week),
  week - last_on_drug_week, 0)
```

This anchors tsd to the last on-drug measurement week, producing `tsd > 0` at the first off-drug timepoint.

Change 4: Restore scalefactor = 2. Add `scalefactor = 2` to the carryover formula in `simulation_clustered.R`:

```
carryover_effect = (1/2)^(2 * tsd / carryover_t_half)
```

Change 5: Time-weighted tod with gap reset. Replace `cumsum(treatment)` with time-weighted accumulation that resets after off-drug gaps:

```
interval = week - lag(week, default = 0)
drug_interval = treatment * interval
# Accumulate only within contiguous on-drug blocks
```

Phase 2: Growth Model Alignment (High)

Change 6: Implement modified Gompertz. Replace the linear rate model with the original's offset-rescaled Gompertz:

```
modgompertz <- function(t, maxr, disp, rate) {
  y <- maxr * exp(-disp * exp(-rate * t))
  vert_offset <- maxr * exp(-disp * exp(-rate * 0))
  y <- y - vert_offset
  y <- y * (maxr / (maxr - vert_offset))
  y
}
```

This requires specifying `max`, `disp`, `rate`, and `sd` parameters for each of the three factors (BR, ER, TR), matching the original's `respparam` data table.

Phase 3: Covariance Refinements (Moderate)

Change 7: Scale ER SDs by expectancy. Multiply ER standard deviations by the expectancy parameter at each timepoint (0.5 for blinded, 1.0 for open-label).

Change 8: Remove BM-ER and BM-TR correlations. Set `Sigma_12[er_idx, 1] = 0` and `Sigma_12[tr_idx, 1] = 0` to match the original, which correlates the biomarker only with BR.

Change 9: Skip BM-BR correlation at timepoint 1. Match the original's `p > 1` guard by setting the first timepoint's biomarker-BR correlation to zero.

Phase 4: Validation and Diagnosis (High)

Change 10: Validation script. Write a script that computes and prints `treatment`, `tod`, `tsd`, `carryover_effect`, and the biomarker-BR correlations for each design/path, comparing against hand-computed reference values from the original.

Change 11: Diagnostic script for power dropoff. For each carryover level (0, 0.1, 0.2):

- Print the full sigma matrix (or at minimum the biomarker-BR correlation subvector)
- Print eigenvalues before and after PD enforcement
- Compare element-wise between `t_half = 0` and `t_half > 0`

Change 12: Regression tests. Add testthat tests for:

- `tsd` values per design/path against reference
- Carryover at boundary conditions
- Null case (biomarker_moderation = 0) rejection rate within binomial bounds of 5%

Phase 5: Extension (the Two-Scenario Payoff)

Change 13: Run baseline comparison. Once Phase 1 is complete, run Scenario 2 (without carryover modeling) and compare power curves against Hendrickson's published heatmaps. This validates the replication.

Change 14: Run two-scenario comparison. Run both scenarios across the full parameter grid. The gap between Scenario 1 and Scenario 2 is the paper's main result: the recoverable power from explicitly modeling carryover.

Change 15: Scale factor investigation. Run with matched scale factors (both 1, both 2) and unmatched (DGP=2, analysis=1) to isolate the scale factor's contribution to power dropoff.

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8. Architectural Decisions to Preserve

These pmsimstats2025 improvements should be kept even when aligning with the original's statistical methodology:

- **Docker + renv + Makefile** for reproducibility
- **furrr parallelism** for runtime efficiency
- **Pre-simulation sigma validation** (eigenvalue checking) – extend to validate per-path sigmas
- **Two-stage conditional MVN** (mathematically equivalent to joint draw, computationally faster)
- **Timestamped output filenames** and **provenance annotations** on figures
- **Structured logging** to file

The AR(1) autocorrelation structure is a deliberate improvement but its impact on power should be quantified. Consider running a sensitivity analysis with compound symmetry (matching the original) vs. AR(1) to isolate this effect.

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9. Open Questions

9.1 Does fixing tsd eliminate excessive power dropoff?

After implementing the interval-based tsd (Phase 1, Change 3), re-run Scenario 2 and compare against Hendrickson's published results. If the power curves match, the tsd bug was the primary cause. If dropoff persists, investigate the scale factor mismatch (Change 15) and PD enforcement distortion (Change 11).

9.2 Is per-path sigma computationally feasible?

The original builds one sigma per path per parameter combination per iteration. With 4 paths and 8 sigma structures (4 designs x 2 c.bm levels), this becomes 32 sigma matrices. Pre-building and caching all 32 is straightforward but requires the Gompertz means (which depend on the response parameters, not just the design geometry). Evaluate whether sigma caching can still work or whether sigma must be rebuilt per parameter set.

9.3 Should tsd_min be a design feature?

The `predictor_comparison.tex` discusses a `tsd_min` workaround (setting a minimum tsd of 0.5 weeks). This is pharmacologically defensible: real trials always have a gap between last dose and assessment. If empirical carryover estimation is pursued, `tsd_min` could become a deliberate design parameter rather than a bug fix. But for replication purposes, it should match the original's interval-based values exactly.

9.4 What is the actual carryover in prazosin trials?

The simulation framework can be applied to real clinical data to estimate actual carryover half-lives. The SUMMARY_WHITE_PAPER.md outlines a research plan for this (Section 12). Empirical estimation would ground the simulation parameters in clinical reality rather than the current hypothetical values.

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10. Document Index

Documents produced during the March 2026 code review:

Document	Scope
`pmsim-audit.md`	Structural inventory of all three repos (Phase 1-5 audit)
`pmsim-power-dropoff-analysis.md`	Variable chain tracing (V1-V5), tsd comparison, recommendations
`pmsim-orig-vs-main-deep-review.md`	Line-by-line DGP comparison (13 differences catalogued)

Key existing documents:

Document	Scope
`predictor_comparison.tex`	tsd bug analysis with numerical tables
`tsd_comparison.md`	Path-by-path tsd values, orig vs. 2025
`hendrickson-carryover-analysis.md`	Discovery: Hendrickson does not model carryover in analysis
`hendrickson-vs-our-detailed-comparison.md`	Post-MVN adjustment analysis, Option 1 recommendation
`why-post-mvn-adjustment-analysis.md`	Correlation vs. additive interaction mechanisms

Document	Scope
pseudocode-comparison-hendrickson-vs-ours.md	Side-by-side pseudocode (9 difference categories)
two-scenario-comparison.md	WITH/WITHOUT carryover modeling experimental design
SUMMARY_WHITE_PAPER.md	High-level framework and research plan
simulation_white_paper.md	Methods paper describing Hendrickson alignment

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11. Summary

pmsimstats2025 has a sound architectural foundation and a well-designed two-scenario comparison framework. However, its data-generating process diverges from Hendrickson's original in five critical ways that prevent meaningful power comparisons:

1. Mean-based vs. correlation-based biomarker interaction
2. Shared vs. per-path sigma matrices
3. Calendar-based vs. interval-based tsd
4. Missing scale factor in carryover formula
5. Count-based vs. time-weighted cumulative drug exposure

These divergences compound: the tsd offset produces 1000x differences in carryover retention, the interaction mechanism produces categorically different statistical signals, and the shared sigma eliminates the treatment-dependent covariance structure that is central to Hendrickson's approach.

The path forward is clear. Phase 1 (Changes 1-5) achieves faithful replication by aligning the DGP with the original. Phase 4 (Changes 10-12) validates the alignment. Phase 5 (Changes 13-15) delivers the scientific payoff: a quantified demonstration that explicit carryover modeling recovers power lost to unmodeled confounding in N-of-1 trial designs.

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